



1. Understand the concept of virtual memory

1.1 Explain and compare how virtual memory is implemented in modern Linux and MS Windows?

Virtual memory is a memory management technique that gives each process its own virtual address space. When RAM does not have enough space to keep all required pages in memory, the operating system can move some pages to secondary storage, such as a disk or SSD, which is much slower than RAM. The operating system and CPU translate virtual addresses into physical RAM addresses using page tables.

The main differences between Linux and Windows are in the operating system level memory management policies and terminology. Linux uses concepts such as swap, page cache, anonymous memory, and memory reclaim, while Windows commonly describes its mechanisms using concepts such as page files, working sets, standby pages, and the Windows Memory Manager.

Although Linux and Windows implement virtual memory differently internally, the fundamental principle is the same. Virtual addresses are translated to physical memory, and the operating system manages which pages are currently resident in RAM and which can be obtained from secondary storage or other backing sources.

1.2 Show step-by-step how to increase or decrease swap space in Linux with screenshots.

First we run “free” to check memory and swap spaces on your computer.

```
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	14Gi	3.9Gi	5.4Gi	91Mi	5.2Gi	10Gi
Swap:	4.0Gi	0B	4.0Gi			

Run “grep swap /etc/fstab” to search for a file with swap in its name which is the swap file

```
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ grep swap /etc/fstab
```

/swap.img	none	swap	sw	0	0
-----------	------	------	----	---	---

Next, run “sudo swapoff /swap.img” to turn off the swap file.

```
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ sudo swapoff /swap.img
[sudo: authenticate] Password:
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	14Gi	4.5Gi	4.5Gi	180Mi	5.4Gi	10Gi
Swap:	0B	0B	0B			

Now, run a series of commands, “sudo fallocate -l 8G /swap.img” to allocate and resize the swapfile to 8GiB, “sudo chmod 600 /swap.img” to make the swap file accessible only to its owner, we do this for security and privacy. Then, “sudo mkswap /swap.img” to format the file and “sudo swapon /swap.img” to activate the swap file.

```
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ sudo chmod 600 /swap.img
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ sudo mkswap /swap.img
mkswap: /swap.img: warning: wiping old swap signature.
Setting up swapspace version 1, size = 8 GiB (8589930496 bytes)
no label, UUID=a732f67b-fd7a-4bf4-aa2f-bf72af7ae4b8
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ sudo swapon /swap.img
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	14Gi	4.4Gi	4.6Gi	180Mi	5.4Gi	10Gi
Swap:	8.0Gi	0B	8.0Gi			

2. Understand the Linux VM Monitoring

2.1 Run “free” command in the Linux shell

2.2 Capture the output and explain everything that is shown by this command

The Command Display

Mem: Memory or RAM.

Swap: Swap space, which is disk space used by the operating system as part of its virtual memory system.

total: The total amount of physical RAM and swap space available in the system.

used: The amount of RAM and swap space currently being used by the system and applications.

free: The amount of RAM and swap space that is currently unused.

shared: The amount of memory being used for shared memory between processes.

buff/cache: The amount of RAM being used for buffers and filesystem cache.

available: The amount of RAM estimated to be available for new applications without needing to use swap.

```
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ free
```

	total	used	free	shared	buff/cache	available
Mem:	15417816	4651804	4785012	179240	5652724	10766012
Swap:	4194300	0	4194300			

2.3 Summarize all numerical values shown by this command into a paragraph.

The computer has a total of 15,417,816 KiB of physical memory RAM. Of this memory, 4,651,804 KiB is currently used, while 4,785,012 KiB is completely free. About 179,240 KiB is used as shared memory, and 5,652,724 KiB is being used for buffers and cache. The system has approximately 10,766,012 KiB of memory available for new applications without needing to use swap. The computer also has 4,194,300 KiB of total swap memory. Currently, 0 KiB of swap is being used, meaning all 4,194,300 KiB of swap remains free.

2.4 Use “free --help” and select 2 useful arguments of the free command

```
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ free -help
free: invalid option -- 'e'

Usage:
  free [options]

Options:
  -b, --bytes          show output in bytes
  --kilo               show output in kilobytes
  --mega              show output in megabytes
  --giga              show output in gigabytes
  --tera              show output in terabytes
  --peta              show output in petabytes
  -k, --kibi          show output in kibibytes
  -m, --mebi          show output in mebibytes
  -g, --gibi          show output in gibibytes
  --tebi              show output in tebibytes
  --pebi              show output in pebibytes
  -h, --human          show human-readable output
  --si                use powers of 1000 not 1024
  -l, --lohi           show detailed low and high memory statistics
  -L, --line           show output on a single line
  -t, --total          show total for RAM + swap
  -v, --committed      show committed memory and commit limit
  -s N, --seconds N    repeat printing every N seconds
  -c N, --count N      repeat printing N times, then exit
  -w, --wide           wide output

  --help              display this help and exit
  -V, --version        output version information and exit
```

could be a combination of multiple arguments if it is interesting. For each argument:

- Describe what it does.
- Show a sample command and explain the output.

```
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ free -h
total        used        free      shared  buff/cache   available
Mem:         14Gi        4.4Gi        4.4Gi       159Mi       5.4Gi       10Gi
Swap:        4.0Gi         0B        4.0Gi

pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ free -h -L
SwapUse      0B  CacheUse    5.4Gi  MemUse      4.4Gi  MemFree     4.4Gi
```

for “-h” it help translate the unit to what is described as human readable output or in the “-L” display the data in one line and only display the swap use memory cache used and memory free

2.5 If you do not have “vmstat”, install and run it in the Linux shell.

2.6 Capture the output and explain everything that is shown by this command

```
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ vmstat
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs  us  sy  id  wa  st  gu
 2   0       0 5430464 106696 4978724    0    0   1024   254 2539    6   4   1 93   2   0   0
```

procs

r: Number of processes waiting to run on the CPU

b: Number of processes blocked.

memory

swpd: The amount of swap space being used.

free: The amount of RAM that is currently unused.

buff: The amount of memory used for kernel buffers

cache: The amount of memory used for page cache.

swap

si: Memory swapped in from disk per second

so: Memory swapped out to disk per second

io

bi: Blocks received from a block device per second

bo: Blocks sent to a block device per second

system

in: Interrupts per second.

cs: Context switches per second

cpu

us: Percentage of CPU time spent running user processes

sy: Percentage of CPU time spent running system (kernel) processes.

id: Percentage of idle CPU time

wa: Percentage of CPU time waiting for I/O

st: Percentage of CPU time stolen by the hypervisor

gu: Percentage of CPU time used to run guest VMs

2.7 Summarize all numerical values shown by this command into a paragraph.

The system currently has 5,430,464 KiB of completely free physical RAM. In addition, 106,696 KiB is used for kernel buffers and 4,978,724 KiB is used as page cache, much of which can be reclaimed if applications require more memory. No swap memory is in use, and there is no swap activity, indicating that RAM is sufficient. CPU usage is also low, with 93% idle time, 4% spent on user processes, 1% on system processes, and only 2% waiting for I/O. Overall, the computer appears to be operating normally without significant CPU, memory, or swap pressure.

2.8 Use “vmstat --help” and select 2 useful arguments of the vmstat command

```
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ vmstat --help
Usage:
  vmstat [options] [delay [count]]

Options:
  -a, --active          active/inactive memory
  -f, --forks           number of forks since boot
  -m, --slabs           slabinfo
  -n, --one-header      do not redisplay header
  -s, --stats           event counter statistics
  -d, --disk            disk statistics
  -D, --disk-sum        summarize disk statistics
  -p, --partition <dev> partition specific statistics
  -S, --unit <char>    define display unit
  -w, --wide           wide output
  -t, --timestamp      show timestamp
  -y, --no-first       skips first line of output

  -h, --help          display this help and exit
  -V, --version       output version information and exit

For more details see vmstat(8).
```

could be a combination of multiple arguments if it is interesting. For each argument:

- Describe what it does.
- Show a sample command and explain the output.

```

pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ vmstat -d
disk- -----reads-----writes-----IO-----
      total merged sectors      ms total merged sectors      ms      cur      sec
loop0      49      0    1356    5567      0      0      0      0      0      4
loop1      14      0      34      6      0      0      0      0      0      0
loop2     6267      0   512874   94186      0      0      0      0      0    32
loop3      56      0    1380    5689      0      0      0      0      0      4
loop4     317      0   20484    4235      0      0      0      0      0      4
loop5      51      0    1350    6445      0      0      0      0      0      6
loop6     513      0   14020   11981      0      0      0      0      0      9
loop7      44      0     706    3644      0      0      0      0      0      3
nvme0n1   1186      0   26220    138      4      0      4      0      0      0
sda       595      0   60568   4425      2      0     16      2      0      3
sdb     25004  16104  6269772 1332554 12397 12753 739562 266602      0    189
loop8      47      0     794   5958      0      0      0      0      0      5
loop9      49      0    1364   5237      0      0      0      0      0      4
loop10     59      0    1416   8040      0      0      0      0      0      6
loop11    2421      0   254294  19277      0      0      0      0      0     16
loop12    104      0    5782   4395      0      0      0      0      0      4
loop13     54      0    1386   8195      0      0      0      0      0      7
loop14    2065      0   98328  30808      0      0      0      0      0     25
loop15    3103      0   27840   5196      0      0      0      0      0      5
loop17    1193      0  115080  13111      0      0      0      0      0     13
loop16     48      0    1350   5911      0      0      0      0      0      5
loop19     84      0    3848   4867      0      0      0      0      0      4
loop18     44      0     702   3780      0      0      0      0      0      3
loop20     44      0     702   3981      0      0      0      0      0      3
loop21     49      0     770   4876      0      0      0      0      0      4
loop22     46      0     706   4508      0      0      0      0      0      4
loop24     21      0     116    175      0      0      0      0      0      0
loop23    813      0   62138  11632      0      0      0      0      0      9
loop25     59      0    1814   1062      0      0      0      0      0      1
loop26     10      0      20      0      0      0      0      0      0      0
pai@pai-H610M-H-DDR4:/run/media/pai/4CE2BF8FE2BF7BB0/KMUTT/Year 3/Operating System$ vmstat -a
procs -----memory-----swap-- -----io-----system-- -----cpu-----
 r  b  swpd  free  inact active  si  so  bi  bo  in  cs  us  sy  id  wa  st  gu
 5   0      0 6929044 2292388 5325776  0   0 2998 296 2189  4  3  1 90  6  0  0

```

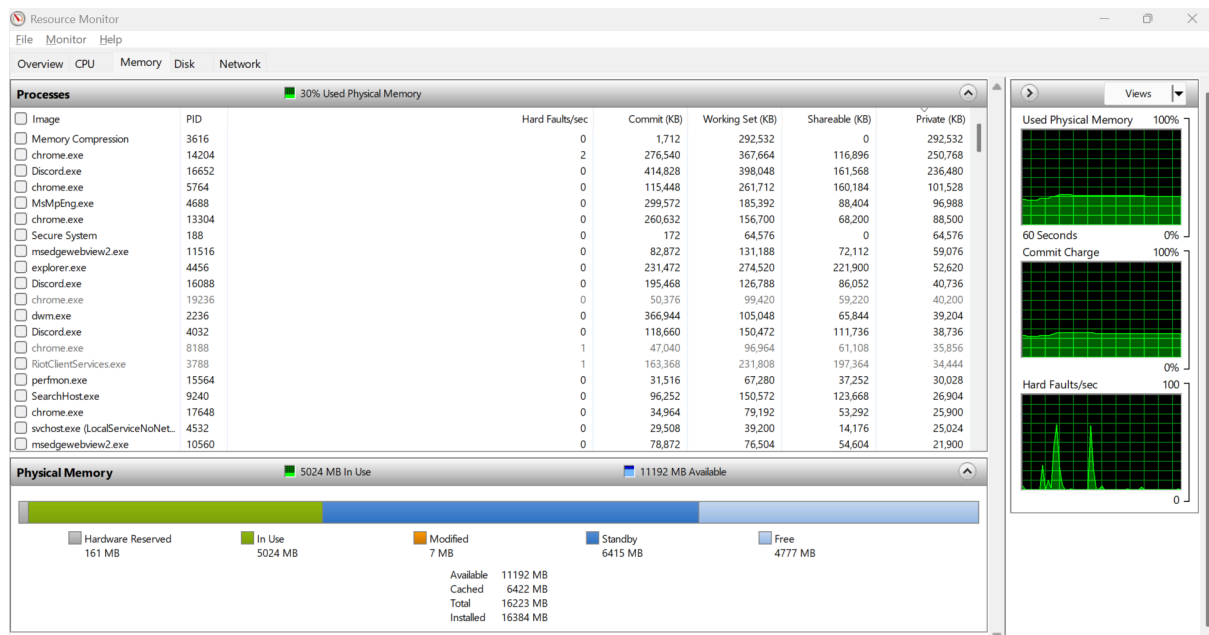
“-a” displays everything mostly similar except that it swaps the buffer and cache with the active and inactive memory.

While, “-d” displays disk I/O statistics for every storage device, showing how much data each device has read and written since the system started. It includes the total number of read and write operations, merged requests, sectors transferred, time spent processing I/O, and current requests in progress.

3. Understand Windows VM Monitoring

3.1 Open “Resource Monitor”

3.2 Capture the output and explain the information shown on the interface related to memory monitoring.



Process

List of programs and background services currently running in Windows.

It Include:

PID

Commit: The amount of virtual memory reserved or committed for a process.

Workset: The amount of physical RAM currently being used by a process. Shareable: The portion of a process's working set that can be shared with other processes, such as shared system libraries or common program files.

Private: The portion of a process's working set that is not shared.

Physical Memory

Hardware Reserved: Memory reserved for hardware by Windows.

In Use: The amount of memory currently used.

Modified: The amount of memory containing changed data that must be written to disk before Windows can reuse it for another purpose.

Standby: The amount of memory that is not actively used by a process but still contains recently used data. Windows can quickly reuse it if needed, or release it when applications require more memory.

Free: The amount of memory that is completely unused.

Available: The amount of memory that can be used if needed to run a new process.

Cached: The amount of memory used for cache.

Total: The total physical memory available for Windows after subtracting hardware-reserved memory.

Installed: The total amount of RAM physically installed in the computer.

View

Used Physical Memory Graph:

Shows the percentage of total physical RAM currently in use.

Commit Charge Graph:

Shows the percentage of committed virtual memory compared with the system's commit limit.

Hard Faults/sec Graph:

Shows the number of hard page faults per second.

3.3 Summarize all numerical values shown in the “physical memory” tab into a paragraph.

The computer has 16,384 MB of installed physical RAM. Of this memory, 161 MB is reserved for hardware, leaving 16,223 MB available for Windows. Currently, 5,024 MB of memory is in use by running applications, system processes, and drivers. About 7 MB is modified memory, while 6,415 MB is in standby memory and 4,777 MB is completely free. The system has 11,192 MB of available memory, which includes standby and free memory that can be allocated to applications when needed. In addition, 6,422 MB is used as cached memory. The Used Physical Memory graph shows that approximately 30% of physical memory is currently in use.

Submission

Submit the document files into LEB2 before the due date. This is the problem session for the group of up to 6 students.